

Lecture 1: Exercises

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MATSE: Operation research

EXERCISE 3.4.9

Ralph loves steaks and potatoes. Therefore, he has decided to go on a steady diet of only these two foods.

Ralph realizes that this isn't the healthiest diet, so he wants to make sure that he eats the right quantities of the two foods to satisfy some key nutritional requirements.

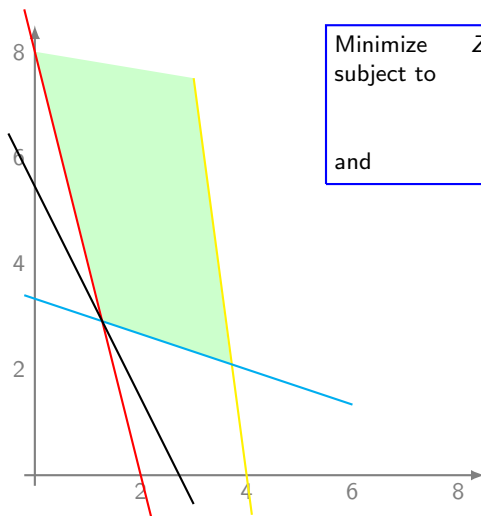
Ralph wishes to determine the number of daily servings of steak and potatoes that will meet these requirements at a minimum cost.

- Formulate a linear programming model for this problem.
- Draw the feasible set
- Find the optimal solution

EXERCISE 3.4.9

Ingredient	Grams of Ingredient per Serving		Daily Requirement (Grams)
	Steak	Potatoes	
Carbohydrates	5	15	≥ 50
Protein	20	5	≥ 40
Fat	15	2	≤ 60
Cost per serving	\$4	\$2	

EXERCISE 3.4.9



Minimize	$Z =$	$4x_S$	$+$	$2x_P$	
subject to		$5x_S$	$+$	$15x_P$	≥ 50
		$20x_S$	$+$	$5x_P$	≥ 40
		$15x_S$	$+$	$2x_P$	≤ 60
and		x_S	$\geq 0,$	x_P	≥ 0

EXERCISE 3.4.14

The Metalco Company desires to blend a new alloy of 40 percent tin, 35 percent zinc, and 25 percent lead from several available alloys with certain properties.

The objective is to determine the proportions of these alloys that should be blended to produce the new alloy at a minimum cost.

- Formulate a linear programming model for this problem.
- Write the problem in matrix form

EXERCISE 3.4.14

Property	Alloy				
	1	2	3	4	5
Percentage of tin	60	25	45	20	50
Percentage of zinc	10	15	45	50	40
Percentage of lead	30	60	10	30	10
Cost (\$/lb)	77	70	88	84	94

EXERCISE 3.4.14

$$\begin{array}{ll}
 \text{minimize} & Z = 77x_1 + 70x_2 + 88x_3 + 84x_4 + 94x_5 \\
 \text{subject to} & 60x_1 + 25x_2 + 45x_3 + 20x_4 + 50x_5 = 40 \\
 & 10x_1 + 15x_2 + 45x_3 + 50x_4 + 40x_5 = 35 \\
 & 30x_1 + 60x_2 + 10x_3 + 30x_4 + 10x_5 = 25 \\
 & x_1 + x_2 + x_3 + x_4 + x_5 = 1 \\
 & x_1, x_2, x_3, x_4, x_5 \geq 0
 \end{array}$$

EXERCISE 3.4.14

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad c = \begin{bmatrix} 77 \\ 70 \\ 88 \\ 84 \\ 94 \end{bmatrix} \quad A = \begin{bmatrix} 60 & 25 & 45 & 20 & 50 \\ 10 & 15 & 45 & 50 & 40 \\ 30 & 60 & 10 & 30 & 10 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 40 \\ 35 \\ 25 \\ 1 \end{bmatrix}$$

EXERCISE 3.1.7

The Whitt Window Company produces:

- Wood-Framed windows (180 euros profit)
- Aluminum-Framed windows (90 euros profit)

There are only 3 employees:

- Doug can make up to 6 wood frames per day
- Linda can make up to 4 aluminum frames per day
- Bob can cut up to 48 square feet of glass per day

The assembly process requires:

- 6 square feet of glass for wood-framed windows
- 8 square feet of glass for aluminum-framed windows

EXERCISE 3.1.7

How many windows to produce of each type to maximize the profit?

- Construct the table
- Formulate the LP
- Find the optimal solution

EXERCISE 3.1.7

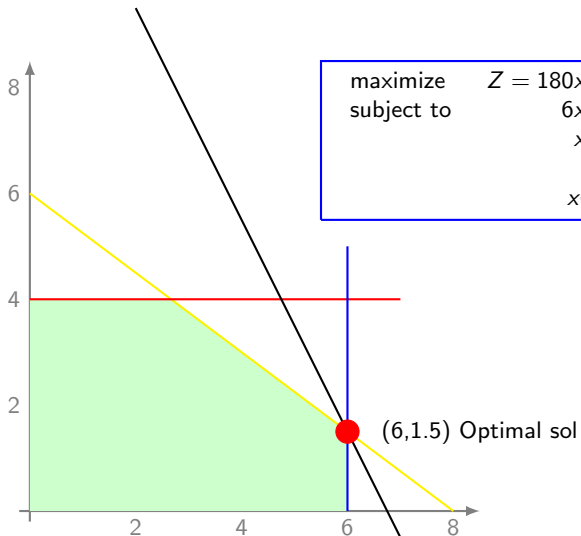
	Resource Usage per Unit of Activity		
Resource	Wood-framed	Aluminum-framed	Available Amount
Wood	1	0	6
Aluminum	0	1	4
Glass	6	8	48
Unit Profit	\$180	\$90	

EXERCISE 3.1.7

	Resource Usage per Unit of Activity		
Resource	Wood-framed	Aluminum-framed	Available Amount
Wood	1	0	6
Aluminum	0	1	4
Glass	6	8	48
Unit Profit	\$180	\$90	

$$\begin{array}{ll} \text{maximize} & Z = 180x_W + 90x_A \\ \text{subject to} & 6x_W + 8x_A \leq 48 \\ & x_W \leq 6 \\ & x_A \leq 4 \\ & x_W, x_A \geq 0 \end{array}$$

EXERCISE 3.1.7

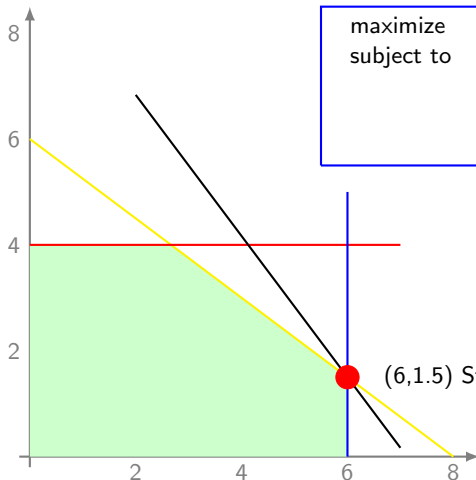


$$\begin{array}{llll} \text{maximize} & Z = 180x_W & + 90x_A & \\ \text{subject to} & 6x_W & + 8x_A & \leq 48 \\ & x_W & & \leq 6 \\ & & x_A & \leq 4 \\ & x_W, & x_A & \geq 0 \end{array}$$

EXERCISE 3.1.7

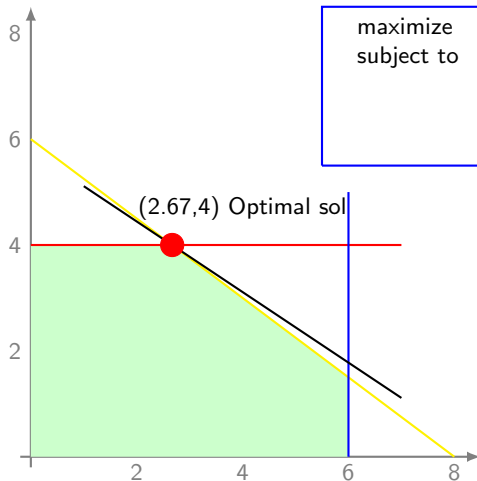
- How would the optimal solution change if the profit for wood-frame windows decrease to 120 euros?
- How would the optimal solution change if the profit for wood-frame windows decrease to 60 euros?
- How does the solution change if Doug can make only 5 wood frames per day?

EXERCISE 3.1.7



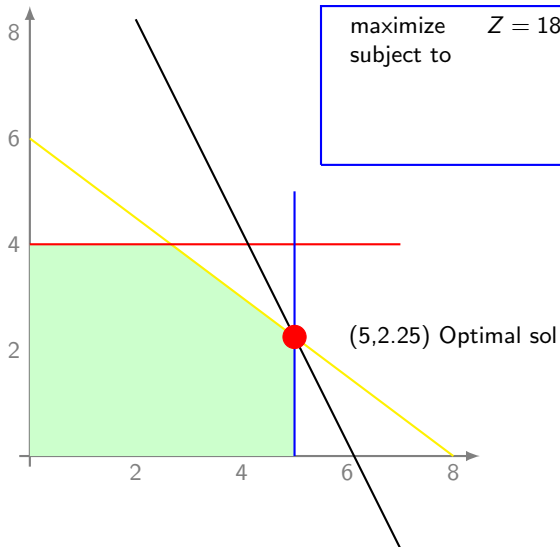
$$\begin{array}{llll} \text{maximize} & Z = 120x_W & + 90x_A & \\ \text{subject to} & 6x_W & + 8x_A & \leq 48 \\ & x_W & & \leq 6 \\ & & x_A & \leq 4 \\ & x_W, & x_A & \geq 0 \end{array}$$

EXERCISE 3.1.7



$$\begin{array}{llll} \text{maximize} & Z = 60x_W & + 90x_A & \\ \text{subject to} & 6x_W & + 8x_A & \leq 48 \\ & x_W & & \leq 6 \\ & & x_A & \leq 4 \\ & x_W, & x_A & \geq 0 \end{array}$$

EXERCISE 3.1.7



$$\begin{array}{llll} \text{maximize} & Z = 180x_W & +90x_A & \\ \text{subject to} & 6x_W & +8x_A & \leq 48 \\ & x_W & & \leq 5 \\ & & x_A & \leq 4 \\ & x_W, & x_A & \geq 0 \end{array}$$